GROMM goes Production

Generative Risk-neutral Option Market Models II - Production Grade

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Abstract

We refine the GROMM model presented in [WBH26] such that it has an expanding strike and expory grid, and its parameters can be estimated based on prevailing grid geometry. We compare LSTM and compact Mamba-style selective state-space backbones for its discrete local-volatility factor law, including variants with state-dependent spot–volatility correlation.

The resulting model has a near-perfect fit to observed option prices and outperforms comparable models in $\mathbb{P}$-measure simulations.

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1 Introduction

2 Factor-Law Evaluation and Portfolio-Conditional Entropic Hedging

2.1 Small-sample predictive evaluation

The historical sample is too short to justify selecting a conditional factor law from its in-sample fit. We therefore use a rolling-origin experiment. The experiment contains four consecutive test blocks of 63 business days, for a total of 252 out-of-fold observations from 13 August 2024 to 28 August 2025. Before each block, a model is fitted using only earlier observations. The number of recurrent training epochs is selected on the final 20% of that earlier sample, after which the model is refitted on the entire pre-block history.

Let $Z_t \in \mathbb{R}^F$ denote the observed factor innovation on day t, and let $C_{m,t}$ be the conditional covariance forecast by model m. The primary $\mathbb{Q}$-measure diagnostic is the Gaussian negative log score per factor and day,

$$L^{\mathbb{Q}}_{m,t} := \frac{1}{2F}\left(\log\det C_{m,t} + Z_t^\top C_{m,t}^{-1} Z_t\right). \tag{1}$$

This is a proper scoring rule for a zero-mean conditional Gaussian factor law. The original models use diagonal $C_{m,t} = \mathrm{diag}(s^2_{m,t,1}, \ldots, s^2_{m,t,F})$. To test state-dependent spot–volatility correlation without estimating an unrestricted loading matrix, we also fit

$$C_{m,t} = D_{m,t}\begin{pmatrix} 1 & \rho_{m,t}^\top \\ \rho_{m,t} & I \end{pmatrix} D_{m,t}, \qquad \rho_{m,t} = \bar\rho \frac{u_{m,t}}{\sqrt{1+\|u_{m,t}\|_2^2}}, \qquad \bar\rho = 0.95, \tag{2}$$

where the first factor is the Gaussianized spot shock and the remaining factors are its unconditionally orthogonal DLV residuals. Hence $\|\rho_{m,t}\|_2 < \bar\rho$ and the covariance is positive definite by construction. Through the fixed DLV loading basis, changing $\rho_{m,t}$ changes both node-level spot–DLV correlations and the effective conditional loading directions.

Performance is reported relative to the AR(1) benchmark as the paired daily improvement

$$\Delta^{\mathbb{Q}}_m := \frac{1}{T}\sum_{t=1}^{T}\left(L^{\mathbb{Q}}_{\mathrm{AR}(1),t} - L^{\mathbb{Q}}_{m,t}\right). \tag{3}$$

Thus, positive values indicate an improvement over AR(1). Confidence intervals resample complete days using a moving-block bootstrap with ten-business-day blocks and 20,000 replications. The conclusions below are unchanged with block lengths of five or twenty business days.

For the physical drift, let $b^{\mathbb{P}}_{t,f}$ denote the estimated coefficient of the attainable $\mathbb{P}$-drift in factor direction f, let $\widehat{b}^{\mathbb{P}}_{m,t,f}$ be its forecast, and let τ_f be its training sample standard deviation. We report

$$L^{\mathbb{P}}_{m,t} := \frac{1}{F}\sum_{f=1}^{F}\left(\frac{b^{\mathbb{P}}_{t,f} - \widehat{b}^{\mathbb{P}}_{m,t,f}}{\tau_f}\right)^2. \tag{4}$$

Unlike (1), this evaluates an estimated attainable drift target. It is not a realized hedging profit-and-loss statistic.

2.1.1 LSTM and Mamba-style selective state-space factor laws

Both recurrent models receive the same causal input x_t: the standardized current DLV-level coordinates concatenated with the previous day's factor innovation. The latter is set to zero whenever the historical panel is not contiguous. From a common hidden state, the output layer predicts the F attainable $\mathbb{P}$-drift coefficients and the F conditional log-scales. The correlated variants additionally predict the $F-1$ unconstrained coordinates u_t in (2). Thus the models differ only in their state update, not in their information set or output parameterization.

For the LSTM, with σ denoting the logistic function, we use

$$(i_t, f_t, g_t, o_t) = \left(\sigma, \sigma, \tanh, \sigma\right)\left(W_x x_t + W_h h_{t-1} + b\right), \tag{5}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t, \tag{6}$$

$$h_t = o_t \odot \tanh(c_t), \qquad y_t = W_o h_t + b_o. \tag{7}$$

The hidden and cell states are reset at a break in the daily sequence.

The model denoted "Mamba" in the experiments is a compact Mamba-style selective state-space recurrence rather than a full stacked Mamba3 network. Its input-dependent retention gate and candidate state are

$$a_t = \sigma(W_a x_t + b_a), \qquad \widetilde{h}_t = \tanh(W_c x_t + b_c), \tag{8}$$

$$h_t = a_t \odot h_{t-1} + (1 - a_t) \odot \widetilde{h}_t, \qquad y_t = W_o h_t + b_o. \tag{9}$$

This has fewer recurrent interactions than the LSTM and provides a useful small-sample test of whether input-dependent memory is sufficient. Both models use 32 hidden coordinates. Their output vector is partitioned as

$$y_t = \left(\widehat{b}_t^{\mathbb{P}}, \ell_t, u_t\right), \qquad s_{t,f} = \bar{s}_f \exp(\mathrm{clip}(\ell_{t,f}, -\log 3, \log 3)), \tag{10}$$

where the final block is omitted in the diagonal models. On each fitting prefix we minimize

$$\frac{1}{T_{\text{train}}} \sum_t L^{\mathbb{Q}}_{m,t} + \frac{1}{T_{\text{train}}} \sum_t L^{\mathbb{P}}_{m,t} + \lambda_\rho \frac{1}{T_{\text{train}}} \sum_t \|\rho_{m,t}\|_2^2, \qquad \lambda_\rho = 10^{-2}, \tag{11}$$

using Adam. The correlation penalty is zero for the diagonal variants. Training duration is selected using only the pre-test validation suffix and the selected number of epochs is then used to refit the complete pre-test history.

Table 1: Rolling-origin factor-law evaluation. Lower mean scores are better. Improvements and confidence intervals are measured relative to AR(1); positive improvements are favourable.

	$\mathbb{Q}$ factor covariance				Attainable $\mathbb{P}$ drift		
Model	Mean score	Improvement	95% interval	Mean $\|\rho_t\|_2$	Mean error	Improvement	95% interval
Fixed	0.5258	−0.0106	[−0.0258, 0.0003]	0	0.7342	−0.3270	[−0.4153, −0.2478]
AR(1)	0.5152	—	—	0	0.4072	—	—
LSTM	0.5063	0.0088	[−0.0471, 0.0658]	0	**0.2722**	**0.1350**	[**0.0993**, **0.1682**]
LSTM with ρ_t	0.6125	−0.0973	[−0.3397, 0.0475]	0.9007	0.2927	0.1145	[0.0663, 0.1572]
Mamba SSM	0.4859	0.0292	[−0.0229, 0.0794]	0	0.3484	0.0588	[0.0100, 0.1023]
Mamba SSM with ρ_t	**0.4796**	**0.0355**	[−0.0476, 0.1096]	0.8682	0.3401	0.0671	[0.0254, 0.1061]

The correlated Mamba-style selective state-space model has the best average $\mathbb{Q}$ score. Relative to its diagonal version, however, its mean improvement is only 0.0063, with a direct paired 95% interval of $[-0.0429, 0.0484]$. The extra correlation state is therefore learnable but does not yet deliver a statistically established predictive gain. The correlated LSTM is materially worse in $\mathbb{Q}$, especially in the third test block, which is direct evidence of overfitting. The LSTM improvement in the attainable $\mathbb{P}$-drift target remains the clearest result. These diagnostics evaluate the dynamics model; final model selection must be based on out-of-sample hedging performance for economically relevant portfolios.

2.1.2 Observable spot–ATM-volatility correlation

The factor score evaluates the latent diffusion, whereas a directly observable leverage diagnostic is the expiry-dependent correlation

$$\rho_{S,\mathrm{ATM}}(\tau) := \mathrm{Corr}\left(\frac{\Delta S_t}{S_{t-1}}, \Delta \log \sigma_{\mathrm{ATM},t}(\tau)\right), \tag{12}$$

where spot returns are forward-adjusted and $\sigma_{\mathrm{ATM},t}(\tau)$ is the Black implied volatility at a fixed rolling business-day tenor τ. The historical curves in Figure 1 use the published SANOS ATM term structure. The full curve contains 890 daily transitions from January 2022 to August 2025; the recent curve contains the 258 transitions from 13 August 2024 onward.

For the model curves, we start from the calibrated state on 29 August 2025 and simulate one business day under $\mathbb{Q}$ using eight DLV substeps and 1,000 antithetic paths. The constant model keeps the unconditional factor loadings, while the Mamba-style selective SSM applies its state-dependent factor amplitudes. At the next date, ATM prices are inverted to Black volatilities at the same rolling tenor. When that tenor lies between two surviving fixed expiries, total variance is interpolated linearly. The 125D simulation point is omitted because a one-day rolling 125D volatility requires the birth of an expiry beyond the terminal model date.

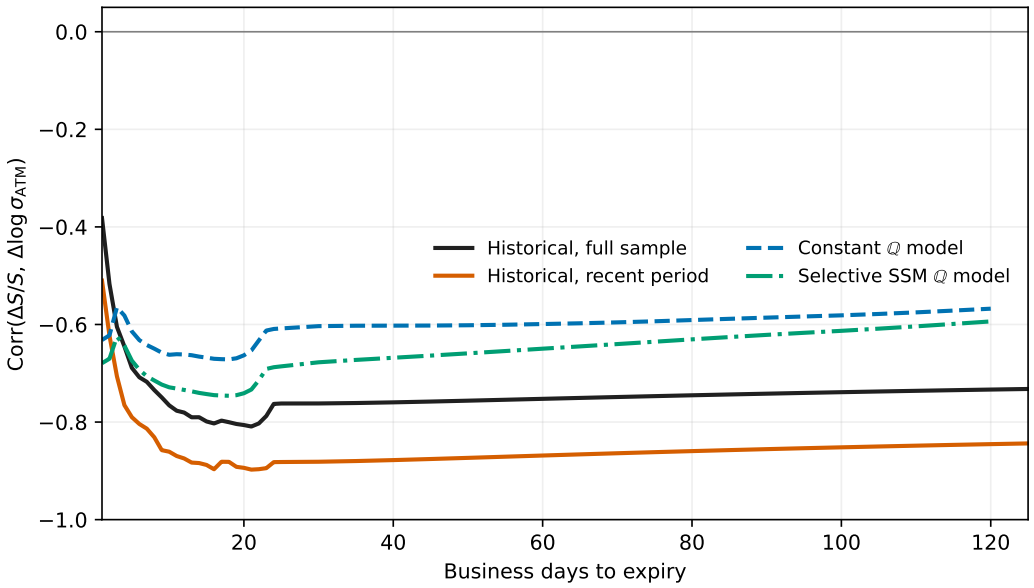


Figure 1: Term structure of daily forward-adjusted spot-return correlation with changes in rolling-tenor ATM Black implied volatility. Historical curves are measured under $\mathbb{P}$; model curves are one-day conditional simulations under $\mathbb{Q}$ from the final calibration state.

Table 2: Selected points from Figure 1.

Business days	Historical	Recent	Constant $\mathbb{Q}$	Mamba SSM $\mathbb{Q}$
1	−0.382	−0.510	−0.632	−0.679
5	−0.689	−0.790	−0.614	−0.674
10	−0.766	−0.861	−0.662	−0.729
20	−0.806	−0.894	−0.663	−0.741
30	−0.762	−0.881	−0.603	−0.678
60	−0.752	−0.869	−0.599	−0.650
90	−0.742	−0.856	−0.586	−0.621
120	−0.733	−0.845	−0.568	−0.594

The selective backbone improves the root-mean-square discrepancy from the full historical

curve from 0.142 to 0.098. It is close to the full-sample 5D correlation (-0.674 versus -0.689), but does not reproduce the stronger recent value of -0.790. More generally, it is too negative at 1–3D and too weak beyond roughly one month. The corresponding discrepancy from the recent curve is 0.185, compared with 0.239 for the constant model. Thus the selective state improves the observable leverage effect, but the maturity shape and recent regime remain material targets for the state-dependent correlation head in Equation (2).

2.2 Portfolio-conditional entropic hedging

Let H be the remaining terminal liability of the current option portfolio, h the vector of current hedge holdings, a the post-trade holdings, R the future gains of the hedge instruments, and $C(a-h)$ the cost of changing the hedge. Conditional on the market state $X_t = x$, define the entropic risk of a residual liability Y by

$$\rho_{\gamma,t}(Y) := \frac{1}{\gamma} \log \mathbb{E}[\exp(\gamma Y) \mid X_t = x], \qquad \gamma > 0. \tag{13}$$

The optimal one-period hedge is

$$a^*_{\gamma,t}(H, h, x) := \arg\min_a \rho_{\gamma,t}\Big(H - a^\top R + C(a-h)\Big). \tag{14}$$

Equation (14) defines one hedge rule for arbitrary portfolios, but not one portfolio-independent hedge position. In general, the rule must be conditioned on the remaining liability H. Current cash wealth does not affect the optimizer because exponential utility has constant absolute risk aversion. Current risky holdings h do matter in the presence of transaction costs, position limits, or other trading frictions.

Without trading costs, an interior optimizer satisfies

$$\mathbb{E}\Big[R \exp\Big(\gamma(H - a^{*\top} R)\Big) \mid X_t = x\Big] = 0. \tag{15}$$

The exponentially tilted measure in Equation (15) depends on H. Consequently, hedges of two portfolios do not generally add, and changing the portfolio notional changes the effective risk aversion. In the frictionless homogeneous case,

$$a^*_{\gamma,t}(cH) = c\, a^*_{c\gamma,t}(H), \qquad c > 0. \tag{16}$$

Thus claim size and risk aversion must be specified together in a hedging benchmark.

There is a useful special case. Suppose that (H, R) is conditionally jointly Gaussian, write $\mu_R := \mathbb{E}[R \mid X_t = x]$, and let $\Sigma_R := \mathrm{Var}(R \mid X_t = x)$ be positive definite. Then

$$a^*_{\gamma,t}(H) = \Sigma_R^{-1} \mathrm{Cov}(R, H \mid X_t = x) + \frac{1}{\gamma} \Sigma_R^{-1} \mu_R. \tag{17}$$

The second term is the position that the optimizer would hold without the liability. Hence the incremental hedge is

$$a^*_{\gamma,t}(H) - a^*_{\gamma,t}(0) = \Sigma_R^{-1} \mathrm{Cov}(R, H \mid X_t = x). \tag{18}$$

It is linear in H and independent of γ. Under $\mathbb{Q}$, where the discounted gains have zero conditional mean, the zero-liability speculative position vanishes. Outside the Gaussian or complete-market setting, Equations (17) and (18) no longer hold in general.

For GROMM, a universal hedging implementation should therefore learn or solve the portfolio-conditioned policy

$$\pi_\gamma(X_t, h_t, \Phi(H)) \,, \tag{19}$$

where $\Phi(H)$ is a representation of the remaining portfolio. A natural choice is its vector of discounted payoffs on common GROMM scenarios, possibly compressed into a payoff basis. The reported hedge is the incremental action

$$\pi_\gamma(X_t, h_t, \Phi(H)) - \pi_\gamma(X_t, h_t, 0) \,. \tag{20}$$

This subtraction removes the speculative position under $\mathbb{P}$ and should be zero under $\mathbb{Q}$ in the absence of costs, constraints, and numerical error.

The ultimate performance statistic is the out-of-sample entropic risk of the realized residual hedge loss. For held-out episodes $p = 1, \ldots, P$, with residual losses Y_p, its empirical value is

$$\widehat{\rho}_\gamma(Y) := \frac{1}{\gamma} \log \left(\frac{1}{P} \sum_{p=1}^{P} \exp(\gamma Y_p) \right) . \tag{21}$$

Comparisons should use identical market episodes, claim notionals, trading costs, position limits, and risk-aversion levels. Factor-law likelihood is a useful dynamics diagnostic, but improvement in Equation (21) is the economically relevant test.

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We want to thank the team at Wharton for their WRDS `https://wrds-www.wharton.upenn.edu` portal and OptionMetrics `https://optionmetrics.com/united-states` for providing their IvyDB data via this service. Without this data our work would not have been possible.

References

[WBH26] Magnus Wiese, Hans Buehler, and Blanka Horvath. GROMM: Generative risk-neutral option market models, 2026. Working paper.